VICTORY 2026

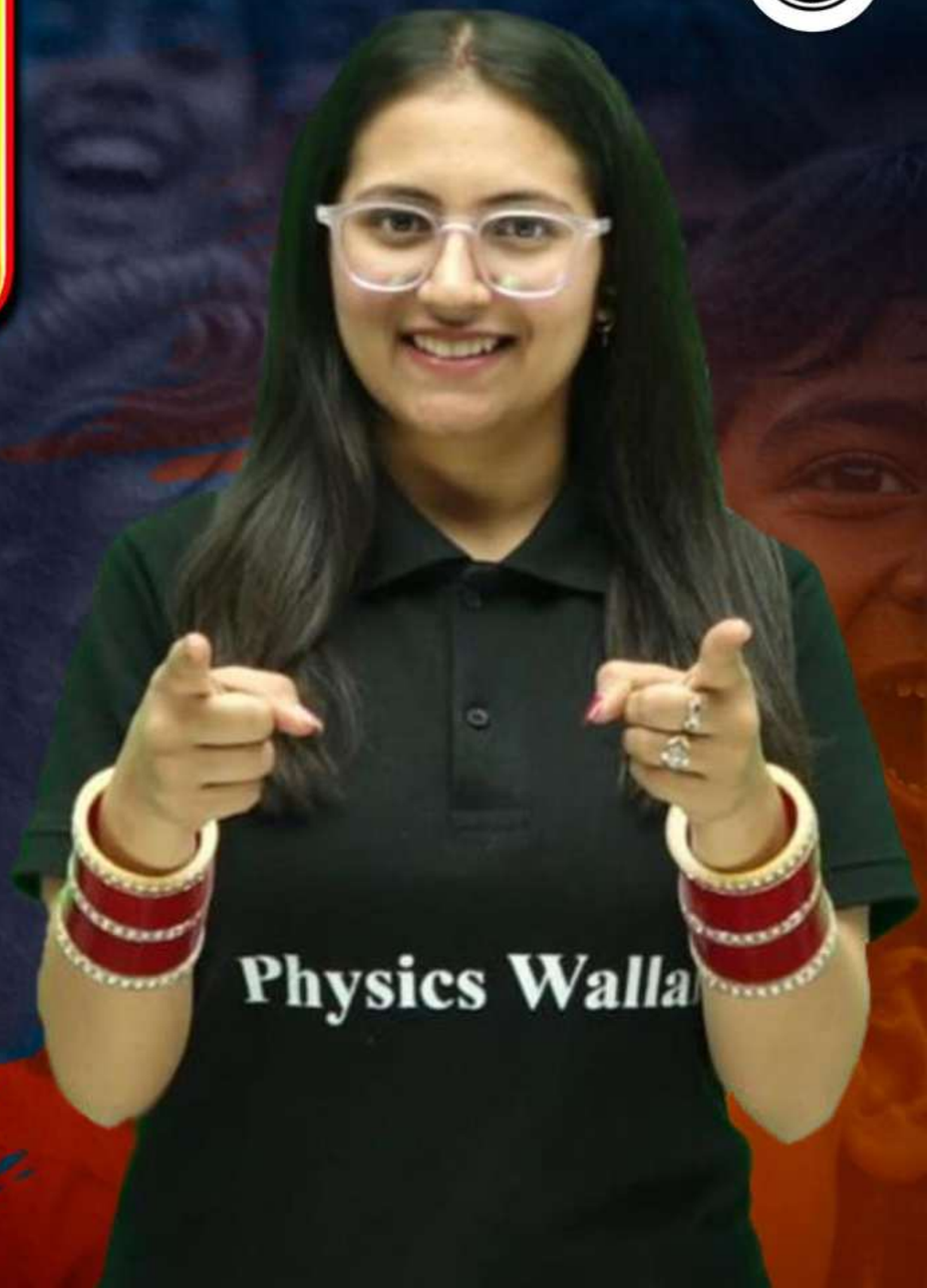
CHEMISTRY

Study of Compounds

(NH_3)

Lect.-03

By- Bharti Ma'am



Topics *to be covered*

A Formation of NH_3

B Chemical and physical properties of NH_3

C Test for NH_3 gas; Uses of NH_3

D Que (MCQ)

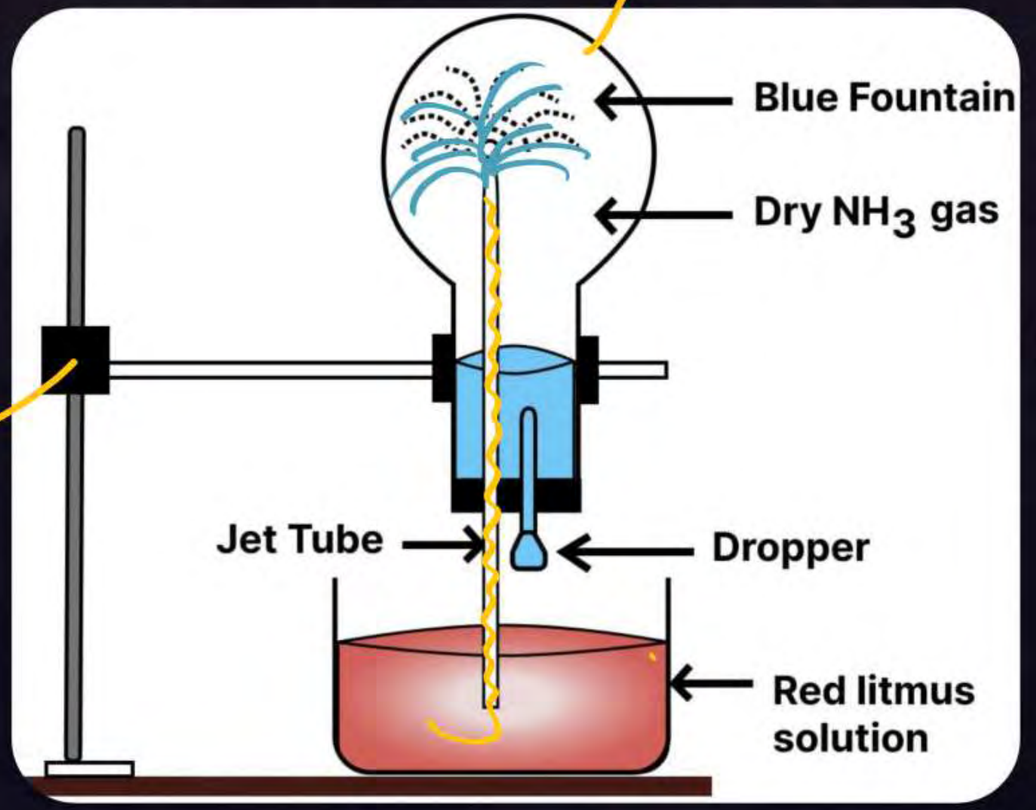
Fountain Experiment

To demonstrate the high solubility of ammonia gas in water.

Apparatus:

1. A round bottomed flask filled with ammonia gas.
2. Mouth of the flask is fitted with a rubber stopper with two holes, one for jet tube and the other for a dropper containing water.
3. A trough containing red litmus solution.
4. A retort stand is used to fix the round bottomed flask in inverted position as shown in Fig..

Fountain Experiment



NH_3 (Basic)

Red Litmus ⟶ Blue

Question

The property exhibited by concentrated sulphuric acid when it is used to prepare hydrogen chloride gas from potassium chloride:

[ICSE 2023]



Note

PYQ

A Dehydrating property

B Drying property

C Oxidizing property

D Non-volatile acid property

$KCl + H_2SO_4 \longrightarrow KHSO_4 + HCl(g)$

Question



The metallic oxide which when reacts with HCl forms salt and water:

[ICSE 2023]

A Carbon monoxide ✗

B Nitrous oxide ✗

C Ammonium hydroxide ✗

D ✓ Sodium oxide Na_2O

Question

Hydrogen chloride gas is not collected over water as: [ICSE 2022]



A. It is highly soluble in water

B. It is less soluble in water

C. It is lighter than air

D. It is heavier than air

Question



Hydrogen chloride can be obtained by adding concentrated sulphuric acid to:

A $NaCl$

B Na_2CO_3

C Na_2SO_4

D Alkali metals

Question

The activity of the Fountain Experiment is to prove that: [ICSE 2016]



A HCl turns blue litmus red

B HCl is denser than air

C HCl is highly soluble in water

D HCl fumes in moist air

Manufacture of ammonia (Haber's Process)

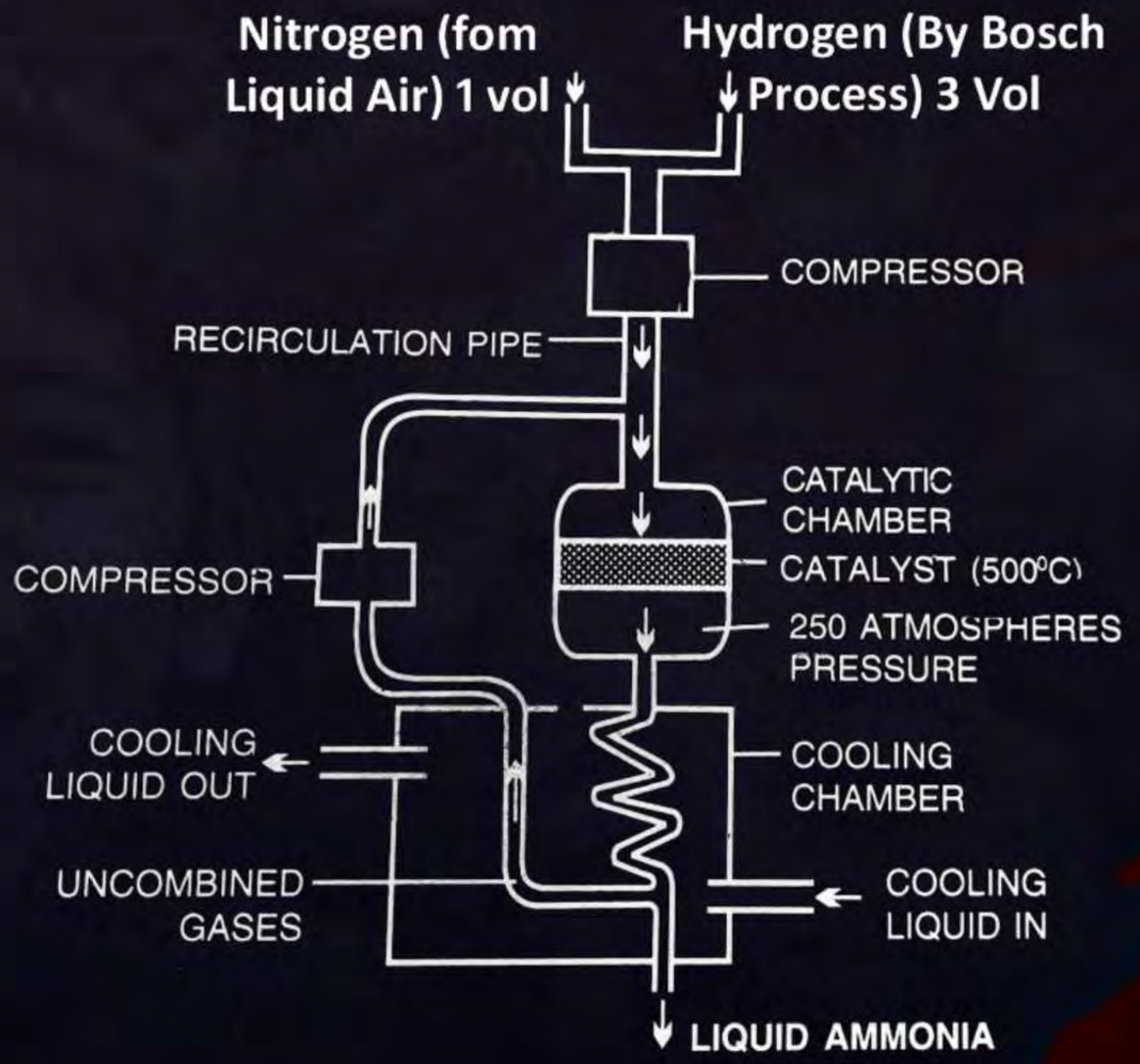
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Manufacture of ammonia by Haber's process

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

By volume $N_2 : H_2$: 1 : 3

* Exothermic process !





Sources of Reactants

Nitrogen gas is obtained by fractional distillation of liquid air.



Process

H_2 gas is obtained from water gas (Bosch process) or from natural gas.

$(Co + H_2)$

Reaction:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + \text{heat.}$$

The reaction is reversible, exothermic and proceeds with decrease in volume.

* Reversible
* exothermic
* Proceeds with dec. in volm



Condition

gmp

Favourable conditions:

Temperature: Optimum temperature is 450 - 500 °C.

Pressure	:	Above 200 atm.
Catalyst	:	Finely divided iron.
Promoter	:	Traces of molybdenum (Mo) or Al_2O_3.
Rate of conversion	:	About 15% of the reacting gases get converted to ammonia

Recovery of Ammonia

Recovery of ammonia:

Ammonia is separated from the unreacted nitrogen and hydrogen by:

1. Liquefaction: Ammonia is liquified easily as compared to nitrogen and hydrogen.

2. Absorbing in water because NH_3 is highly soluble in water, while N_2 and H_2 are very slightly soluble



Sources of Reactants

1. Nitrogen and hydrogen are dried, purified and then mixed in 1: 3 respectively, at a relatively high pressure. (Above 200 atm)

2. This mixture is passed in an electrically heated catalytic chamber finely divided iron with aluminium oxide or a little amount of molybdenum at a temperature of 500°C.



Sources of Reactants

3. Mixture of ammonia formed along with residual nitrogen and exchanges their heat with the incoming unreacted nitrogen and hydrogen mixture.

4. The hot mixture of the outgoing gases contains nearly 15% ammonia. The mixture is passed through the cooling pipes (condenser).

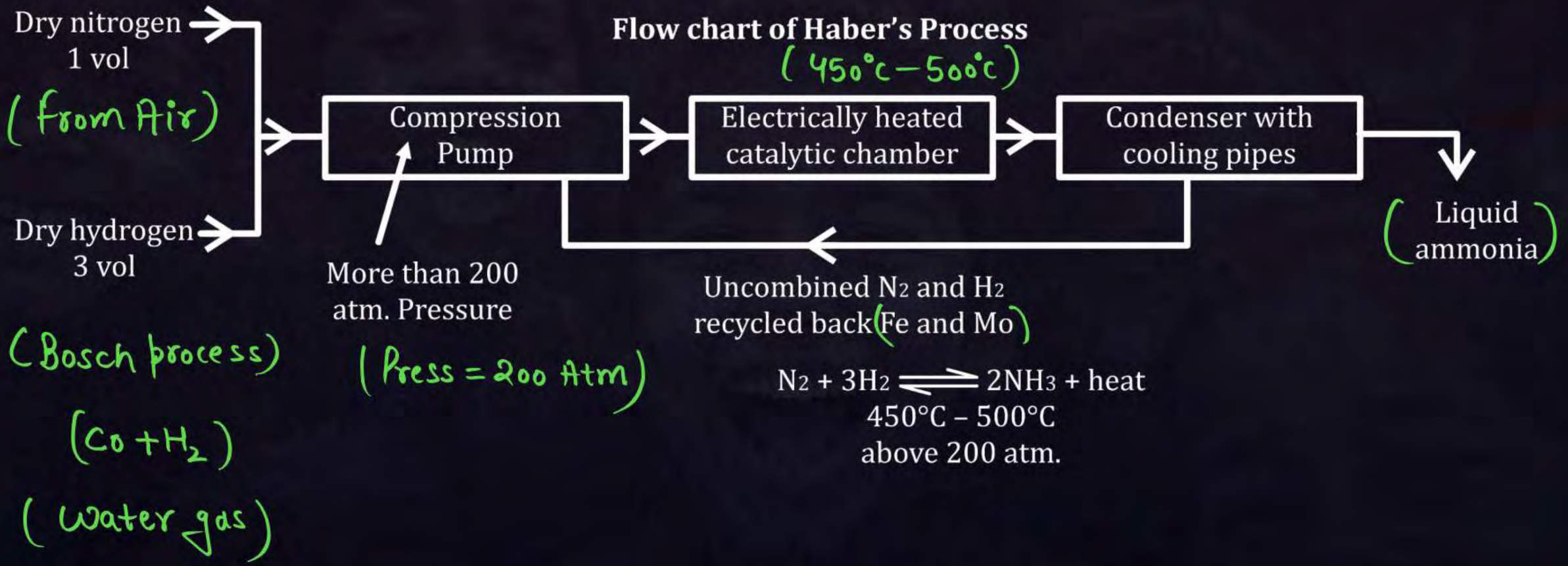


Sources of Reactants

5. Ammonia liquefies first, whereas nitrogen and hydrogen do not liquefy easily

6. The unchanged nitrogen and hydrogen are recirculated through the plant to get more ammonia.

Flow chart of Haber's Process





Physical properties of ammonia

1. Colour : Colourless.
2. Odour : Strong, pungent choking smell.
3. Taste : Slightly bitter (alkaline) taste.
4. Physiological action : Non-poisonous. If inhaled, affects the respiratory system and brings tears to eyes. It is a powerful heart stimulant and may cause death when inhaled in large quantities.
5. Density : Vapour density = 8.5 [Lighter than air]



Physical properties of ammonia

7. Liquefaction : Easily liquefied at 10°C by compressing it at 6 atm. pressure.

8. Boiling point : Liquid ammonia boils at -33.5°C.

9. Freezing point : Solid NH_3 melts at -77.7°C.

10. Solubility : Highly soluble in water, 1 vol. of water dissolves about 702 vols. of ammonia at 20°C and 1 atm. pressure

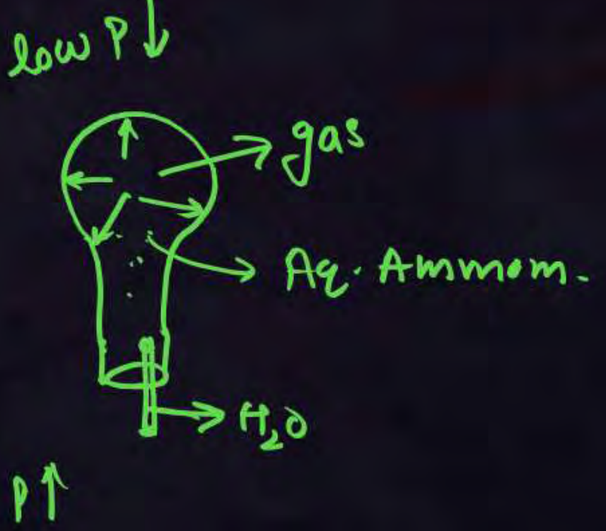
Chemical properties of ammonia

V. Imp

1. **Thermal dissociation:** Ammonia gas dissociates into nitrogen and hydrogen at high temperature or by electric sparks.

$$2NH_3 \overset{\Delta}{\rightleftharpoons} 3N_2 + 3H_2$$

(g)



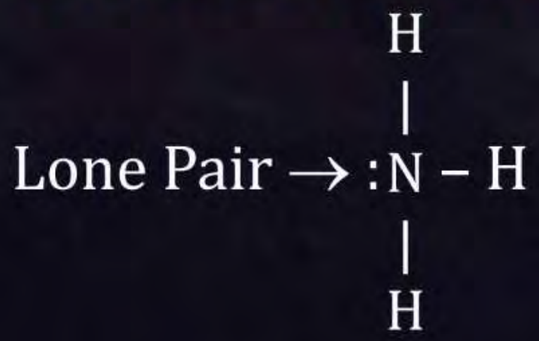
$$2NH_3 \overset{\Delta}{\rightleftharpoons} N_2 + 3H_2$$

Chemical properties of ammonia

2. **Basic Properties:** Dry ammonia is a covalent molecule. So, it is neutral even in liquefied form.

Imp.

Its aqueous solution is a weak base, the basicity is due to a lone pair of electrons on its nitrogen atom.



Lewis Base :→ e^- donate



Chemical properties of ammonia

- An aqueous solution of ammonia [NH_4OH] is a weak base. It dissociates partially to give hydroxyl ions.

- The alkaline behaviour is due to the hydroxyl ions [OH^-]

$$NH_3 + H_2O \rightleftharpoons 2NH_4OH$$

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

Chemical properties of ammonia

$NH_3 \rightarrow$ Basic gas

Action on indicators

Indicator	Change in colour	
	Original	To
Moist litmus	Red	Blue
Methyl orange	Orange	Yellow
Phenolphthalein	Colourless	Pink



Chemical properties of ammonia

3. **Burning of ammonia in oxygen :** Ammonia is not a supporter of combustion. It extinguishes a burning splint and does not burn in air. It burns in oxygen with greenish yellow flame.

MCQ

Chemical properties of ammonia

4. **Catalytic oxidation of ammonia:** In the presence of platinum at 800° C, ammonia reacts with oxygen to give nitric oxide and water vapour.

$$NH_3 + O_2 \xrightarrow{800^\circ C} NO + H_2O$$

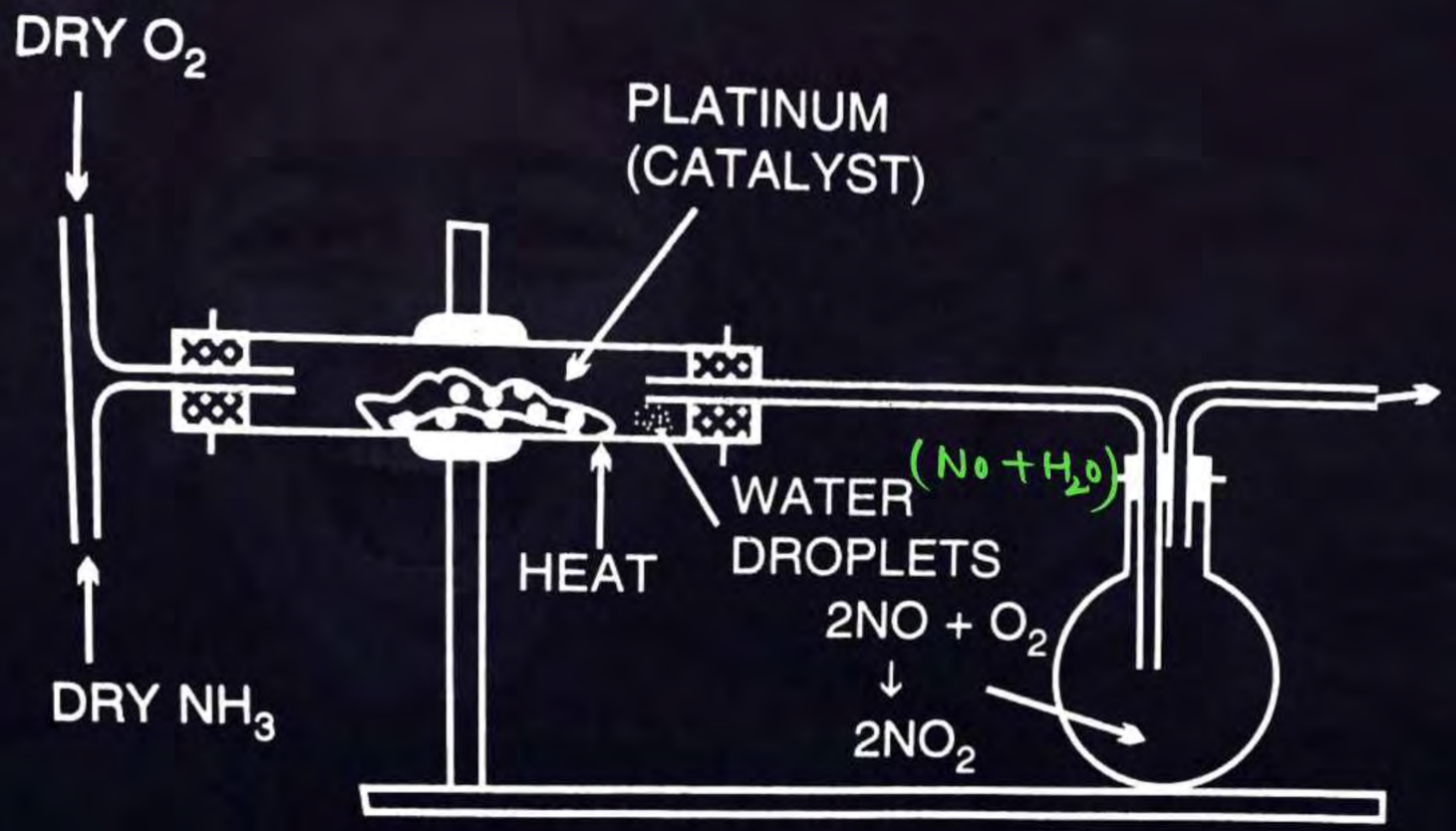
$$NO + O_2 \longrightarrow NO_2$$

(Nitric oxide)

Chemical properties of ammonia

Procedure:

Pass dry ammonia gas and oxygen gas through inlets over heated platinum placed in



Ammonia as a reducing agent

(i) Reduction of metallic oxides

Ammonia reduces heated metallic oxides to give Metals, water vapour and nitrogen.

Procedure:

Pass ammonia gas over heated copper oxide in a hard glass tube.

Reaction:

$$2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2\uparrow$$

Black solid — reddish brown solid — gas

Removal of O_2 : Redⁿ

NH_3: Reducing Agent

Removal of O_2

Ammonia gas (Act as Reducing Agent) + Metal oxide → Metal + H_2O + $N_2\uparrow$

Ammonia as a reducing agent

Observation: The black copper oxide is reduced to reddish brown copper.

CuO (Black)

Similarly, ammonia reduces heated yellow lead monoxide to greyish metallic lead.

R.A

$$3PbO + 2NH_3 \rightarrow 3Pb + 3H_2O + N_2\uparrow$$

yellow greyish

Removal of oxygen (Red^n)

Cu (Reddish Brown)

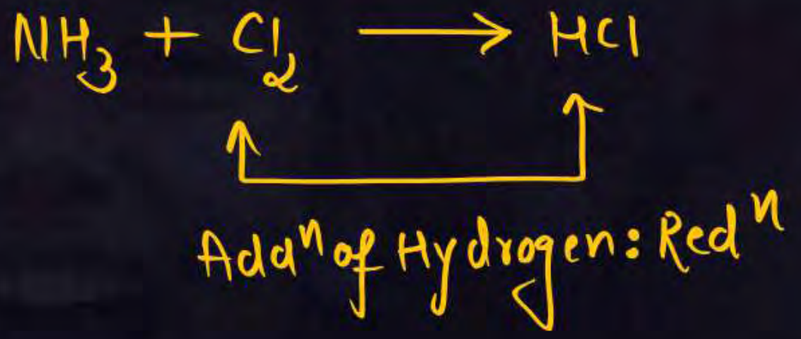
Ammonia as a reducing agent

ii) Reduction of chlorine to hydrogen chloride:

When chlorine reacts with ammonia, it is reduced to HCl, which further combines with excess of ammonia and forms NH_4Cl.

a) When ammonia is in excess.

(as Reducing Agent)



Ammonia as a reducing agent

Observation

The greenish-yellow colour of chlorine disappears and white fumes of ammonium chloride are formed.

b) When chlorine is in excess, then the products are hydrogen chloride and yellow coloured highly explosive liquid nitrogen trichloride.

$$NH_3 + 3Cl_2 \rightarrow 3HCl + NCl_3$$

→ Highly explosive liq.



Ammonia as a reducing agent

These reactions show that:

- Chlorine has great affinity for hydrogen.

- Ammonia is a reducing agent

Ammonia as a reducing agent

6. Reaction with acids

Ammonia being a weak base accepts protons [hydrogen ions] to form ammonium ions.

It reacts with acids to form ammonium salts.

Ammonia gas	Acid	Ammonia salt
NH_3 (gas) +	HCl (gas) →	NH_4Cl (solid) Ammonium chloride
NH_3 (gas) +	HNO_3 →	NH_4NO_3 Ammonium nitrate
$2NH_3$(gas) +	H_2SO_4 →	$(NH_4)_2SO_4$ Ammonium sulphate



Ammonia as a reducing agent

Ammonia is dissolved in water to form its aqueous solution (ammonium hydroxide).

$$NH_3 + H_2O \rightarrow NH_4OH$$

Ammonium hydroxide reacts with acids to form salt and water.



Ammonia as a reducing agent

Ammonia + [aq.]	Acid →	Ammonium + salt	Water
NH_4OH +	$HCl \rightarrow$	NH_4Cl +	H_2O
NH_4OH +	$HNO_3 \rightarrow$	NH_4NO_3 +	H_2O
$2NH_4OH$ +	$H_2SO_4 \rightarrow$	$(NH_4)_2SO_4$ +	$2H_2O$

Ammonia as a reducing agent

7. **Reaction of aqueous solution of ammonia with soluble metal salts:** Ammonium hydroxide reacts with soluble salts of metals to give insoluble precipitates of their respective metallic hydroxides, (except sodium and potassium).

This behaviour is similar to NaOH solution and KOH solution.

In Acid-Base and salt NH_4OH



Test

Ammonia gas can easily be recognized by following characteristics:

1. It has a sharp characteristic odour.

2. It turns
   - Moist red litmus blue,
   - Moist turmeric paper brown
   - Phenolphthalein solution pink.

Test

Ammonia gas can easily be recognized by following characteristics:

3. It gives dense white fumes with conc, hydrochloric acid.
   $NH_3 + HCl \rightarrow NH_4Cl$. (White fumes)

$NH_3(g) + CuSO_4 \longrightarrow$ Blue ppt $\xrightarrow[NH_3 \text{ (excess)}]{+}$ 'deep Blue'

4. When NH_3 gas is passed through copper sulphate solution blue precipitate is formed which changes to deep blue solution on passing the gas in excess.



Test

Ammonia gas can easily be recognized by following characteristics:

5. All ammonium salts generally produce ammonia gas when warmed with alkali like sodium hydroxide.

6. Ammonium salt gives brown colour or precipitates with Nessler's solution K_2HgI_4 (a slightly alkaline solution of potassium mercuric iodide made by adding KOH).

K_2HgI_4 gmf

Uses of Ammonia

1. Liquid ammonia is used as a refrigerant in ice plants:

a) Liquid ammonia takes a lot of energy to vaporize. This heat is taken from the surrounding bodies which are consequently cooled down.

b) Anhydrous ammonia is a clear, colourless liquid under pressure. It evaporates rapidly and produces cooling effect. This makes ammonia a good refrigerant.



Uses of Ammonia

1. Liquid ammonia is used as a refrigerant in ice plants:

c) The main refrigerants used are freon chlorofluro carbon (CFC). They deplete ozone layer and also contribute to global warming.



Uses of Ammonia

V. Imp PYQ

Advantages of ammonia as refrigerant

i) Ammonia is environmentally compatible. It does not deplete ozone layer and docs not contribute towards the global warming.



Uses of Ammonia

Advantages of ammonia as refrigerant

ii) It has superior thermodynamic qualities as a result ammonia refrigeration systems use less electricity.



Uses of Ammonia

Advantages of ammonia as refrigerant

iii) Ammonia has a recognizable odour and so leaks are not likely to escape. It being lighter than air goes up in the atmosphere not affecting the life too much on earth.



Uses of Ammonia

2. Aqueous NH_3 emulsifies or dissolves fats, grease, etc. so it is used.

i) To remove grease and perspiration stains from clothes

ii) For cleaning tiles, windows.



Uses of Ammonia

3. Ammonia is used in the manufacture of:

a) Nitrogenous fertilisers like ammonium sulphate, diammonium hydrogen phosphate, ammonium nitrate and urea.

Ammonia reacts with carbon dioxide at 150°C and 150 atm. pressure to give urea which is a valuable nitrogenous fertiliser.

$$2NH_3 + CO_2 \xrightarrow[150\ atm]{150^\circ C} \underset{\text{urea}}{NH_2CONH_2} + H_2O$$



Uses of Ammonia

b) Explosives like ammonium nitrate. Ammonium nitrate decomposes explosively leaving no residue behind

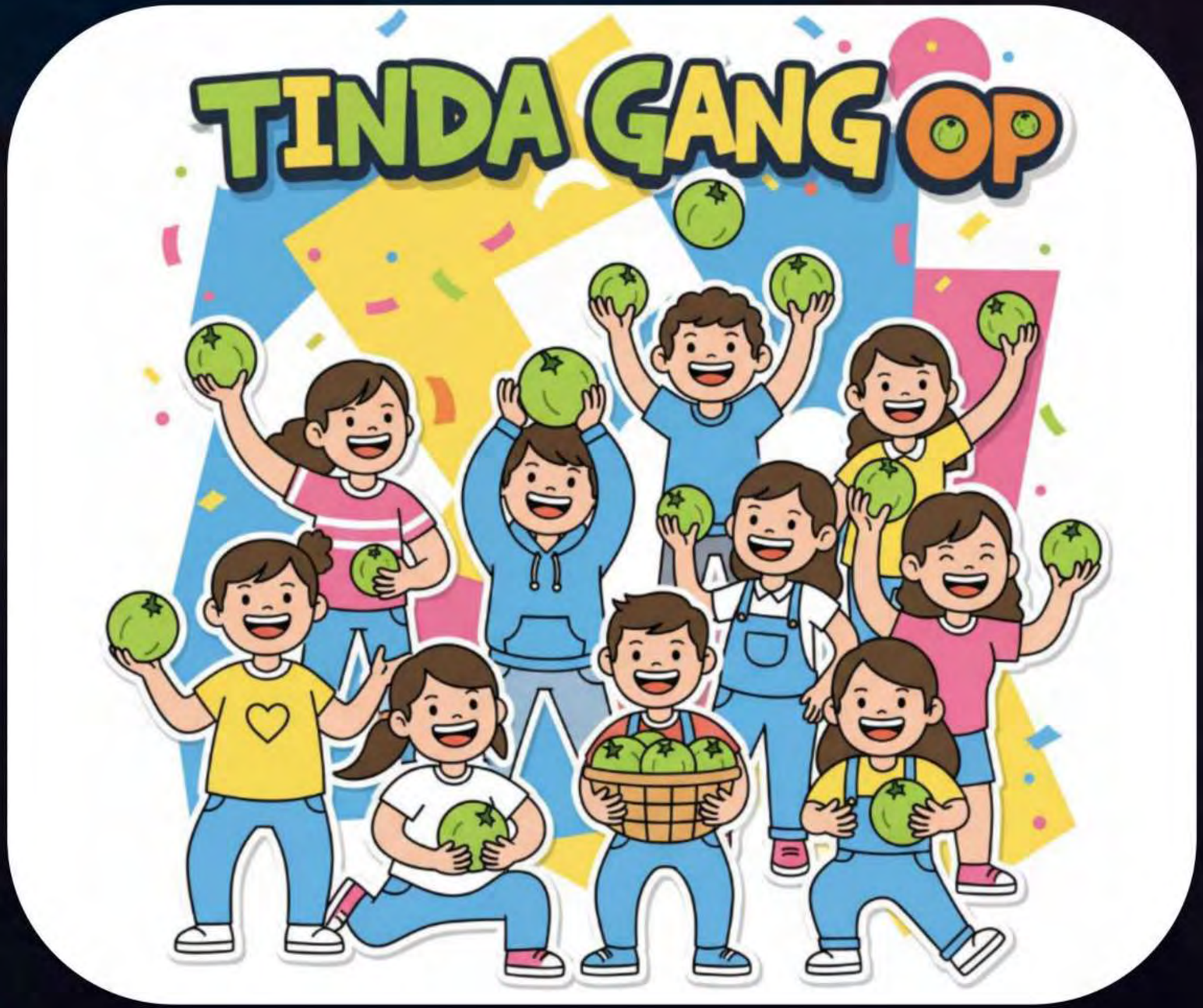
$$NH_4NO_3 \xrightarrow{\Delta} N_2O(g) + 2H_2O(vap.)$$



Uses of Ammonia

c) Sodium carbonate by Solvay process. This process involves the interaction of ammonia.


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